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Zeus, the future of decentralized climate forecasting

Subnet 18 takes a fundamentally different path on weather modelling, small location-specific models outperforming centralised SOTA by ~40% on RMSE, with miners competing region by region.

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The global climate has become increasingly unpredictable due to violent Show more weather events, with every industry imaginable affected at a large scale by increasingly difficult-to-predict weather patterns. According to Terms of Service | Privacy Policy SBloomberg, in 2024, even hedge funds increased their talent pool of Accessibility | Ads info | More

weather experts by 23%.

As climate volatility intensifies, accurate and reliable climate forecasting is imperative for multiple industries, from agriculture and transportation to energy management and retail. Traditional weather prediction systems

are built into complex models that demand high levels of compute.

Built on the SBittensor blockchain, SZeusSubnet introduces a novel method for weather forecasting that leans into the principles of decentralization, open markets, and competition among machine learning models. The project replaces centralized, compute-heavy models with an incentive-driven system of contributors, each vying to produce the most accurate local forecasts.

SZeusSubnet operates on Bittensor's Subnet 18 with three key actors:

1. Miners: Run localized machine learning models, providing weather predictions for specific geographies and timeframes - the best model is named the Best Recent Performer (BRP). The validator proxy routes forecasts through the BRP.

2. Validators: Evaluate the accuracy of miner forecasts against real- world outcomes using sampled environmental data from ERA5 (a global atmospheric reanalysis dataset produced by the SCopernicusECMWF Climate Change Service (C3S) at the European and assign a difficulty-adjusted reward
3. Subnet Owner: Oversees governance and incentive mechanisms that keep the network aligned and competitive. Validates using heavier models like Aurora (by SMSFTResearch).

Zeus was launched in early February, initially deploying on the Bittensor testnet and releasing their whitepaper, highlighting their vision, architecture and incentive design. Led by co-founders @wouterhar, @0xtravvv and their team, the project aims to enable “decentralized weather forecasting in a controlled environment”. The subnet went live on Mainnet on March 24, 2025, with initial results exceeding expectations: by June 5, Miners were outperforming SOTA (state-of-the- art), consistently achieving superior forecasting accuracy compared to benchmarks set by centralized models.

While other weather-modeling based subnet projects rely on one model, SGaia_AI_ using SMSFTResearch's Aurora, for example, Zeus takes a fundamentally different path. Instead of scaling one model to cover the globe, Zeus splits the problem into smaller, location-specific tasks.

This approach has a clear advantage: local miners, focused on narrow geographies like small cities or regions, can optimize lightweight models that outperform SOTA baselines with less compute. According to a paper released by the team, Zeus miners using the BRP (Best Recent Performer) achieved an RMSE (Root Mean Square Error) of 1.05K, a 39.8% improvement over the centralized baseline of 1.74K. Since then, Zeus released that the best miner for the 2 day forecast window for windspeed achieved an average RMSE of 0.89 across its window, 45.6% better than the baseline. The team also focuses on a difficulty-weighted RMSE - predicting in more stable areas will pay less than accurate predictions in complex regions.

Zeus’ model has again evolved, using both on- and off-the-grid data. Zeus’ model has again evolved, using both on- and off-the-grid data.

On-the-grid global data, including values derived from modelling, and off-the-grid data, including direct observations from weather stations, create a set of what Zeus calls “hybrid data”, working to enhance accuracy and reliability.

Forecasting isn't just about accuracy, it’s about incentives. Zeus uses SBittensor’s token economy to reward contributors for performance. Engineers compete in a decentralized “mixture-of-experts” setup, where validators assess which miners are delivering the most accurate results in each micro-region.

The result is a self-improving system: miners iterate on one another’s models, validators drive constant benchmarking, and subnet owners shape the rules of incentives. The project plans to scale beyond temperature and precipitation, dewpoint and windspeed to other critical climate variables, with potential implications for green energy pricing.

Zeus’ architecture is tailored for wide-scale commercialization. With API already deployed, the team delivers localized, real-time forecasts directly to the markets that depend on them. Zeus’ APIs have widespread use-case, from decentralized prediction platforms like SKalshi and SPolymarket to insurance, logistics, and agricultural sectors.

The team’s long-term vision is to grow their fully monetizable climate forecasting stack providing incentivized, low-cost, scalable solutions, governed by an open community. By focusing on out-of-sample validation and minimizing data overhead for validators, Zeus offers a much leaner alternative to legacy systems.

Just last month - on September 19 - Zeus launched its official API, with four tiers of access:

• $0/month - free plan with 1000 API calls per day, all weather parameters, global coverage and 7-day forecasts

• $40/month - “Developer” plan with 10,000 API calls per day, low latency and technical support

• $1000/month - “Business” plan with 1,000,000 API calls per day, priority technical support and early access to probabilistic forecasts

• And a custom “Enterprise” plan with unlimited API calls, dedicated technical support, custom integrations and on-premises deployment options

The team is working to make the response time even faster for customers, with 30% of miner score based on speed of answer, with response times from the API nearing real-time.

Zeus’ team also teased that once their 14-day prediction option becomes available, the price may be adjusted, but early adopters will get to keep their price. The now-launched API is targeting a client base of energy traders, hedge funds, drone operators, retailers, and weather- warning systems.

Before launch, the team shared that 50% of the revenue from the API will go towards Alpha token “buy backs” to reward miners and alpha token holders.

To showcase the accuracy of their API, in a recent SOpenTensor Novelty Search, founder Jacob Steeves (@const_reborn) asked the Zeus team to use the API to predict the maximum temperature for the next day in Seoul. Zeus’ team predicted a maximum temperature for the day after to be 26 degrees Celcius, comparing their prediction to Google’s 28 degree Celcius forecast.

A day later, Zeus' prediction from the day before (26 degrees Celcius) matched the temperature, making Google's weather prediction of 28 degrees Celcius a whole two degrees off, showcasing the level of unmatched accuracy this new modeling type produces. Zeus | SN 18 @zeussubnet · Sep 26, 2025 Yesterday, during @opentensor Novelty Search, @const_reborn asked us to demo Zeus live:

"Max temperature in Seoul tomorrow?" Zeus: 26°C Google: 28°C

Today? It was 26°C. Live, verified, beat SGoogse. No better validation than that.

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SGAIA_AI_ (Subnet 57) in comparison, works to fine tune the existing Microsoft Aurora model. Zeus does not specifically use Aurora for anything other than validating because they believe the results and models are only slightly better but cost millions of dollars for slight increases in performance over time.

In late August, Zeus announced its first subnet partnership with SN41, Sportstensor, to provide the weather intelligence layer for SSportstensor's sports betting intelligence app, Almanac.

At the end of August, Zeus introduced a new version of Dendrite to the subnet, helping speed up transmission of validator challenges to miners by four seconds. In improving the speed at which validators transmit new challenges to miners, the team cut the delay between the first and last miner to just 0.1 seconds. Not only did they implement the change, but provide the release open-source, available on the Zeus GitHub.

Zeus | SN 18 @zeussubnet · Aug 26, 2025 Bittensor subnets can face major networking issues. Here's our fix. Subnet owners, pay attention.

Most subnets, including Zeus, use the standard Dendrite to send challenges from validators to miners. For subnets handling lots of data/querying many miners, it's slow: up to 5 Show more

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Earlier this week, Zeus began benchmarking against SECMWF IFS HRES (High Resolution Integrated Forecast System from the European Centre for Medium-Range Weather Forecasts) and Microsoft Aurora, signalling the team's readiness to not only validate the subnet's output quality, but to address doubts of the model "head-on". Forecasts sourced from IFS to address doubts of the model "head-on". Forecasts sourced from IFS HRES are integrated into the validator's code, while the team will integrate the Aurora model themselves. The results come out on Friday, October 10, so only a few days will tell how Zeus' forecasting measures

up.

The roadmap for Zeus includes expanded forecasting variables, finer geographic resolution, and tighter validator feedback loops. There are also plans to integrate real-time economic feedback, allowing forecasts to

dynamically adjust based on shifting grid demand or agricultural trends.

Looking ahead, the team expects to cut forecast error across all variables by over 30% in the next 24 months. As more miners specialize and validators refine performance metrics, Zeus is positioning itself as a decentralized oracle for one of the most important datasets in the world: climate prediction.

To showcase their model, Zeus will be launching an app using the subnet's API. Zeus' goal is for customized apps to integrate with the API, creating tailored solutions for every industry.

As climate change accelerates, the need for a robust system of forecasting tools makes Zeus a premier global asset. Zeus promotes transparency and democratization of a field that has traditionally been private and restricted.

Zeus is shifting global climate forecasting from centralization to an adaptive, dynamic, incentivized ecosystem that, if successful, will produce more accessible and reliable forecasts by incentivizing engineers to develop and optimize forecasting models, and minimizing storage and hardware requirements for validators.

By turning weather prediction into a decentralized, incentivized competition, Zeus is disrupting an industry long dominated by government agencies and billion-dollar tech firms. If it succeeds, it won't just forecast the weather, it'll reshape the infrastructure behind how we prepare for it.

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